
Andrei Draganescu
**Algebraic Multigrid Preconditioning of Distributed
Optimal Control Problems Constrained by Elliptic
Equations Based on Smoothed Aggregation**

Department of Mathematics and Statistics
University of Maryland Baltimore County
1000 Hilltop Circle
Baltimore MD 21250
`draga@umbc.edu`

In this work we propose a new algebraic multigrid (AMG) method for the reduced form of linear-quadratic optimal control problems constrained by elliptic equations of the form

$$\min_{y,u} \frac{1}{2} \|y - y_d\|^2 + \frac{\beta}{2} \|u\|^2, \quad -\Delta y = u. \quad (1)$$

Previous work has shown that problems like (1) can be solved very efficiently using multigrid methods, assuming a geometric hierarchy of grids is available. As with geometric multigrid, AMG methods have been originally designed for solving linear systems representing discrete differential equations, where the matrices are sparse. Instead, the linear systems arising in the reduced formulation of (1), the so-called reduced Hessians, are not sparse, therefore AMG methods are not directly applicable. Relying on the basis functions and prolongation operators originating in smoothed aggregation for the elliptic equation, we construct an AMG-based multigrid preconditioner for (1) that exhibits an efficiency similar to the geometric multigrid case, namely it increases with increasing resolution.